Linear Algebra 1

Adar's Study Series

LINEAR ALGEBRA 1

Systems of Linear Equations · Echelon Forms and General Solutions
Vectors, Linear Combinations and Span · The Matrix Equation $A\vec{x} = \vec{b}$
Solution Sets · Linear Independence · Linear Transformations

A First Course in Linear Algebra

Deep Violet & Amber Edition

Adar's Study Series

Linear Algebra 1

Preface

This book is a complete first course in linear algebra. It begins where the subject begins — with systems of linear equations and the row operations that solve them — and develops, step by step, the central ideas of a first semester: echelon and reduced row echelon forms, pivots and free variables, vectors and their geometry, linear combinations and span, the matrix equation $A\vec{x} = \vec{b}$, the structure of solution sets, linear independence, and linear transformations with their standard matrices.

How the book is organised.

- **Chapter 1** introduces linear systems, the augmented matrix, elementary row operations, and the goal of row reduction.
- **Chapter 2** develops echelon form (REF) and reduced row echelon form (RREF), pivots and free variables, a general two-pass reduction method, and the three consistency cases.
- **Chapter 3** turns to vectors: the spaces $\mathbb{R}^n$, vector operations and their geometry, linear combinations, span, and the equivalence between systems, vector equations, and span membership.
- **Chapter 4** studies the matrix equation $A\vec{x} = \vec{b}$: how $A\vec{x}$ is computed, and how a system is written in vector and matrix form.
- **Chapter 5** classifies solution sets: homogeneous and nonhomogeneous systems, the solution structure theorem, and the geometric picture in $\mathbb{R}^3$.
- **Chapter 6** develops linear independence and dependence, the pivot test, and conceptual reasoning with dependent sets.
- **Chapter 7** presents linear transformations: matrices as functions, the two defining properties of linearity, the standard matrix, projections and rotations, and the onto / one-to-one classification with Rank–Nullity reasoning.

A note on numbering. The chapters correspond to Parts 1–7 of the study-series notes: Chapter k corresponds to Part k (the linear-independence notes, labelled "Part 5" in their source material, appear here as Part 6, so that the sequence of parts remains unbroken and Chapter 7 follows as announced).

How to use this book. Every chapter contains worked examples carried out in full, highlighted answers, remarks that flag common pitfalls, and a summary of key points. The reader should work through the examples with pencil and paper; row reduction is learned by doing. No background beyond school algebra is assumed.

— Adar's Study Series

Notation

- $\mathbb{R},\ \mathbb{R}^2,\ \mathbb{R}^3,\ \mathbb{R}^n,\ \mathbb{R}^m$ — the real numbers and their n-fold products.
- Vectors are written with arrows: $\vec{x},\ \vec{b},\ \vec{v}_1,\ \vec{e}_1,\dots,\vec{e}_n,\ \vec{0}$; the $\vec{e}_i$ are the standard basis vectors.
- $[\,A \mid \vec{b}\,]$ — the augmented matrix of a linear system; the vertical bar separates coefficients from constants.
- Elementary row operations: $R_i \to kR_i$ (scaling), $R_j \to R_j \pm kR_i$ (replacement), $R_i \leftrightarrow R_j$ (interchange).
- In pattern matrices: $\blacksquare$ denotes any non-zero entry, $*$ any number, and 0 a zero entry.
- $\mathrm{Span}\{\vec{v}_1,\dots,\vec{v}_k\}$ — the set of all linear combinations of the given vectors.
- $T_A : \mathbb{R}^n \to \mathbb{R}^m$ — the linear transformation associated with A; also $\vec{x} \mapsto A\vec{x}$.
- $\Rightarrow$, $\Leftrightarrow$, $\Longleftrightarrow$, $\Updownarrow$ — implies; equivalent to.

Contents

Chapter 1

Linear Equations and Systems

Part 1 · Foundations

1.1. What Is Linear Algebra?

Linear Algebra is the branch of mathematics that studies linear equations, systems of linear equations, vectors, matrices, and linear transformations.

Main Goal

Find the values of unknown variables that satisfy one or more linear equations simultaneously.

Applications

- Artificial Intelligence (AI)
- Machine Learning (ML)
- Data Science
- Computer Graphics
- Computer Vision
- Signal Processing
- Engineering

1.2. Systems of Linear Equations

Definition 1.1 (Linear equation)**:** A *linear equation* in the variables $x_1, \dots, x_n$ is an equation that can be written in the form $a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$, where all a_i and b are real constants. Equivalently: each variable appears only to the first power, only multiplied by a constant, and variables are never multiplied together. Equations containing terms such as x_1x_2, x_1^2, $\sqrt{x_1}$, or $\sin x_1$ are *not* linear.

Definition 1.2 (System of linear equations)**:** A **system of linear equations** is a collection of two or more linear equations involving the same variables.

General Form

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

1.3. Solving by Hand: A System in Two Variables

Example 1.1 (Substitution and elimination)**:** Find x and y such that:

$$x - y = -2 \tag{1}$$
$$3x - 6y = -15 \tag{2}$$

Method 1 — Substitution

From equation (1):

$$x - y = -2 \quad \Rightarrow \quad x = y - 2.$$

Substitute into equation (2):

$$3(y-2) - 6y = -15 \quad \Rightarrow \quad 3y - 6 - 6y = -15 \quad \Rightarrow \quad -3y = -9 \quad \Rightarrow \quad y = 3.$$

Then:

$$x = y - 2 = 3 - 2 = 1.$$

Answer: (x, y) = (1, 3)

Remark 1.1: *Limitation: the substitution method becomes messy as the number of variables and equations grows.*

Method 2 — Elimination

Multiply equation (1) by 3:

$$3x - 3y = -6.$$

Subtract from equation (2):

$$(3x - 6y) - (3x - 3y) = -15 - (-6) \quad \Rightarrow \quad -3y = -9 \quad \Rightarrow \quad y = 3.$$

Substitute back into equation (1):

$$x - 3 = -2 \quad \Rightarrow \quad x = 1.$$

Answer: (x, y) = (1, 3)

Geometric Interpretation

Each equation represents a straight line in the 2D plane.

- Intersecting lines $\rightarrow$ One unique solution
- Parallel lines $\rightarrow$ No solution
- Same line $\rightarrow$ Infinitely many solutions

For Example 1.1, the two lines intersect at exactly one point:

Intersection Point: (1, 3)

1.4. Three Variables: A System with Infinitely Many Solutions

Example 1.2: Find x, y and z such that:

$$x - 2y + 4z = 1 \tag{1}$$
$$x + \ \ y + 7z = 3 \tag{2}$$

Solving the System

Subtract equation (1) from equation (2):

$$(x + y + 7z) - (x - 2y + 4z) = 3 - 1 \quad \Rightarrow \quad 3y + 3z = 2 \quad \Rightarrow \quad y = \frac{2}{3} - z.$$

Substitute into equation (1):

$$x - 2\left(\tfrac{2}{3} - z\right) + 4z = 1 \quad \Rightarrow \quad x = 1 + \tfrac{4}{3} - 2z - 4z \quad \Rightarrow \quad x = \frac{7}{3} - 6z.$$

Let the free variable $z = t$ (t is a parameter):

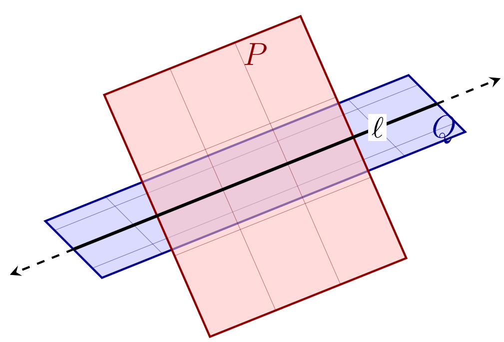
$$x = \frac{7}{3} - 6t, \qquad y = \frac{2}{3} - t \quad \textbf{and} \quad z = t \qquad (t \in \mathbb{R})$$

Geometric Interpretation

Each of the two equations represents a **plane** in 3-dimensional space:

- Plane P: all points (x, y, z) satisfying $x - 2y + 4z = 1$;
- Plane Q: all points (x, y, z) satisfying $x + y + 7z = 3$.

The planes P and Q are **not parallel**. Therefore they **intersect each other in a straight line** ℓ — not in a single point, and not nowhere. Every point of ℓ satisfies *both* equations at once, so the solution set of the system is exactly the line ℓ: infinitely many solutions.



The plane P passes through the plane Q; they intersect in the straight line ℓ (dashed parts indicate that the planes and the line extend infinitely).

Conclusion: the system has **infinitely many solutions**. As the parameter t runs through $\mathbb{R}$, the point $\left(\frac{7}{3} - 6t,\ \frac{2}{3} - t,\ t\right)$ traces out exactly the line ℓ of intersection of the two planes.

1.5. Linear Equations in n Variables

General Form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

Where:

- $x_1, x_2, \ldots, x_n \rightarrow$ Variables
- $a_1, a_2, \ldots, a_n \rightarrow$ Coefficients
- $b \rightarrow$ Constant

General Linear System

$$\begin{aligned}
a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\
a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\
&\vdots \\
a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m
\end{aligned}$$

1.6. The Augmented Matrix

Definition 1.3 (Augmented matrix)**:** The **augmented matrix** packages every coefficient and constant of a linear system into a single array, separating the coefficients from the constants with a vertical bar:

$$\left[\begin{array}{cccc|c}
a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\
a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\
\vdots & \vdots & & \vdots & \vdots \\
a_{m1} & a_{m2} & \cdots & a_{mn} & b_m
\end{array}\right]$$

- **Left part** $\rightarrow$ Coefficient Matrix
- **Right part** $\rightarrow$ Constant Column

Goal of Row Reduction

Reduce the augmented matrix using elementary row operations until the solution of the system can be read off directly.

1.7. Elementary Row Operations

1. **Row Scaling:** $R_i \rightarrow kR_i \quad (k \neq 0)$. Multiply every entry of a row by a non-zero constant.
2. **Row Replacement:** $R_j \rightarrow R_j \pm kR_i$. Add/subtract a multiple of one row to another row.
3. **Row Interchange:** $R_i \leftrightarrow R_j$. Swap the positions of two rows.

Key Fact: elementary row operations never change the solution set of the system — they only change its appearance.

1.8. Row Reduction in Action

Example 1.1 revisited via the augmented matrix

Starting augmented matrix:

$$\left[\begin{array}{cc|c} 1 & -1 & -2 \\ 3 & -6 & -15 \end{array}\right]$$

Apply the operation $-3R_1 + R_2 \to R_2$:

$$\left[\begin{array}{cc|c} 1 & -1 & -2 \\ 0 & -3 & -9 \end{array}\right]$$

From Row 2: $-3y = -9 \;\Rightarrow\; y = 3$. From Row 1: $x - y = -2 \;\Rightarrow\; x = 1$.

Answer: (x, y) = (1, 3)

Example 1.2 revisited via the augmented matrix

Starting augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & -2 & 4 & 1 \\ 1 & 1 & 7 & 3 \end{array}\right]$$

Apply the operation $R_2 - R_1 \to R_2$:

$$\left[\begin{array}{ccc|c} 1 & -2 & 4 & 1 \\ 0 & 3 & 3 & 2 \end{array}\right]$$

From Row 2: $3y + 3z = 2 \;\Rightarrow\; y = \frac{2}{3} - z$. Substitute the expression of y obtained from Row 2 into Row 1:

$$x = \frac{7}{3} - 6z.$$

Let $z = t$ (free parameter):

$$x = \frac{7}{3} - 6t, \qquad y = \frac{2}{3} - t \quad \textbf{and} \quad z = t \qquad (t \in \mathbb{R})$$

1.9. Looking Ahead: Echelon Form

Learning Goals

- Learn Echelon Form (EF) and Reduced Row Echelon Form (RREF).
- Solve linear systems using systematic Row Reduction.
- Determine the number of solutions of a linear system.

Types of Solutions

Inconsistent System

- No Solution

Consistent System

- One Unique Solution
- Infinitely Many Solutions

1.10. Summary

- Linear Algebra studies linear equations and systems of linear equations.
- An augmented matrix packages coefficients and constants into one array.
- Row reduction simplifies a system without changing its solution set.
- Three elementary row operations (Definition **??**): Row Scaling, Row Replacement, Row Interchange.
- Echelon Form (EF) and Reduced Row Echelon Form (RREF) are standard forms used to solve systems efficiently.
- A linear system may have: No Solution, One Unique Solution, or Infinitely Many Solutions.

With row reduction in hand, the next chapter develops echelon form and reduced row echelon form in depth.

Chapter 2

Echelon Forms and General Solutions

Part 2 · REF, RREF, Pivots & Consistency Cases

2.1. Echelon Form

Definition 2.1 (Echelon form)**:** A matrix is in **echelon form** if it satisfies the following two conditions:

- All non-zero rows are above any **zero rows**.
- Each non-zero **leading entry** (the first non-zero number from the left in a row) is to the **left** of the non-zero leading entries of the rows below it.

Notation Key

- $\blacksquare$ or 1 $=$ any non-zero number
- $*$ $=$ any number
- 0 $=$ zero entry

Examples of Echelon Form

A 2×3 example:

$$\begin{bmatrix} \blacksquare & * & * \\ 0 & 0 & \blacksquare \end{bmatrix}$$

A 4×6 example:

$$\begin{bmatrix} 0 & \blacksquare & * & * & * & * \\ 0 & 0 & 0 & \blacksquare & * & * \\ 0 & 0 & 0 & 0 & 0 & \blacksquare \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

2.2. Reduced Row Echelon Form

Definition 2.2 (Reduced row echelon form)**:** A matrix is in **reduced row echelon form** if it meets the criteria for echelon form (Definition 2.1), and additionally satisfies:

- Every non-zero leading entry (**pivot**) equals 1.
- There are zeros above each non-zero leading entry (pivot), as well as below it.

Examples of RREF

A 2×3 example:

$$\begin{bmatrix} 1 & * & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad \text{A } 4 \times 6 \text{ example:} \quad \begin{bmatrix} 0 & 1 & * & 0 & * & 0 \\ 0 & 0 & 0 & 1 & * & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

2.3. Two Key Theorems

Theorem 2.1 (Uniqueness of RREF)**:** *Any matrix can be transformed into a unique reduced row echelon form using elementary row operations.*

Theorem 2.2 (Invariance)**:** *Elementary row operations never change the solution set of the underlying linear system.*

2.4. Pivots and Free Variables

Once a matrix is in echelon form, the following terms describe its structure:

- **Pivot** $\rightarrow$ The non-zero leading entry of a row.
- **Pivot Column** $\rightarrow$ A column that contains a pivot (non-zero leading entry).
- **Pivot Variable** $\rightarrow$ The variable corresponding to a pivot column.
- **Free Column** $\rightarrow$ A non-pivot column in the coefficient matrix (excluding the augmented column).
- **Free Variable** $\rightarrow$ The variable corresponding to a non-pivot column.

Remark 2.1: *Note for augmented columns: the final column of an augmented matrix (the right-hand constants) is **NOT** a free column — it does not belong to the coefficient matrix or correspond to any variable.*

2.5. The Reduction Method: Two Passes

Step 1 — Forward Pass (top-left to bottom-right)

Work from top-left to bottom-right, using elementary row operations to bring the matrix into Echelon Form (REF).

Step 2 — Backward Pass (bottom-right to top-left)

Work from bottom-right to top-left to bring the matrix into Reduced Row Echelon Form (RREF): work backward to create zeros above each pivot and scale pivots to 1.

2.6. Worked Example: Full Reduction to RREF

Example 2.1: Put the following matrix into reduced row echelon form, then find the general solution.

$$\left[\begin{array}{ccccc|c} 0 & 0 & 1 & 4 & -1 & 1 \\ 2 & -2 & 4 & 6 & 3 & 7 \\ 1 & -1 & 1 & -1 & 2 & 1 \end{array}\right]$$

Part (a) — Row Operations to RREF

Forward Pass (to Echelon Form)

Swap R_1 and R_3 $(R_1 \leftrightarrow R_3)$:

$$\left[\begin{array}{ccccc|c} 1 & -1 & 1 & -1 & 2 & 1 \\ 2 & -2 & 4 & 6 & 3 & 7 \\ 0 & 0 & 1 & 4 & -1 & 1 \end{array}\right]$$

Eliminate R_2's leading entry $(-2R_1 + R_2 \to R_2)$:

$$\left[\begin{array}{ccccc|c} 1 & -1 & 1 & -1 & 2 & 1 \\ 0 & 0 & 2 & 8 & -1 & 5 \\ 0 & 0 & 1 & 4 & -1 & 1 \end{array}\right]$$

Swap R_2 and R_3 $(R_2 \leftrightarrow R_3)$:

$$\left[\begin{array}{ccccc|c} 1 & -1 & 1 & -1 & 2 & 1 \\ 0 & 0 & 1 & 4 & -1 & 1 \\ 0 & 0 & 2 & 8 & -1 & 5 \end{array}\right]$$

Eliminate the entry below the pivot $(R_3 - 2R_2 \to R_3)$ — this is the Echelon Form:

$$\left[\begin{array}{ccccc|c} 1 & -1 & 1 & -1 & 2 & 1 \\ 0 & 0 & 1 & 4 & -1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 3 \end{array}\right]$$

Backward Pass (to RREF)

Clear the entries above the pivot in Column 5 $(R_2 + R_3 \to R_2)$ and $(R_1 - 2R_3 \to R_1)$:

$$\left[\begin{array}{ccccc|c} 1 & -1 & 1 & -1 & 0 & -5 \\ 0 & 0 & 1 & 4 & 0 & 4 \\ 0 & 0 & 0 & 0 & 1 & 3 \end{array}\right]$$

Clear the entry above the pivot in Column 3 $(R_1 - R_2 \to R_1)$:

$$\left[\begin{array}{ccccc|c} 1 & -1 & 0 & -5 & 0 & -9 \\ 0 & 0 & 1 & 4 & 0 & 4 \\ 0 & 0 & 0 & 0 & 1 & 3 \end{array}\right]$$

2.7. Reading Off the General Solution

Continuing Example 2.1, from the RREF augmented matrix:

$$\left[\begin{array}{ccccc|c} 1 & -1 & 0 & -5 & 0 & -9 \\ 0 & 0 & 1 & 4 & 0 & 4 \\ 0 & 0 & 0 & 0 & 1 & 3 \end{array}\right]$$

Identifying the Variables

- **Pivot Columns:** $1, 3, 5 \;\rightarrow\;$ **Pivot Variables:** x_1, x_3, x_5
- **Free Columns:** $2, 4 \;\rightarrow\;$ **Free Variables:** x_2, x_4

System Equations

$$\begin{aligned} x_1 - x_2 - 5x_4 &= -9 \\ x_3 + 4x_4 &= 4 \\ x_5 &= 3 \end{aligned}$$

General Solution

$$\begin{aligned} x_1 &= x_2 + 5x_4 - 9 \\ x_2 &= \textbf{free} \\ x_3 &= -4x_4 + 4 \\ x_4 &= \textbf{free} \\ x_5 &= 3 \end{aligned}$$

2.8. Consistent vs. Inconsistent Systems

- A **consistent system** is a linear system that has at least one solution (unique or infinitely many).
- An **inconsistent system** is a linear system that has no solutions.

Remark 2.2: *When reducing an augmented matrix $[\,A \mid \vec{b}\,]$, exactly one of three cases will apply — covered in the next three sections.*

2.9. Case 1: No Solution

Condition

- The last (constant) column contains a pivot.
- **Equivalently:** a row of the form $[\,0\ 0\ \cdots\ 0 \mid \blacksquare\,]$ exists, with $\blacksquare \neq 0$.

Reason: such a row translates to the equation $0 = \blacksquare$, which is logically impossible.

Example

$$\left[\begin{array}{ccc|c} 1 & 0 & 7 & -12 \\ 0 & 0 & 1 & 7 \\ 0 & 0 & 0 & 5 \end{array}\right] \quad \Longrightarrow \quad \begin{aligned} x_1 + 7x_3 &= -12 \\ x_3 &= 7 \\ 0 &= 5 \ \leftarrow \ \textbf{Impossible!} \end{aligned}$$

Conclusion: No Solution (Inconsistent)

2.10. Case 2: A Unique Solution

Condition

- The last column is **NOT** a pivot column, **AND**
- There are **NO** free variables — every coefficient column has a pivot.

Example

$$\left[\begin{array}{cc|c} 1 & -1 & 4 \\ 0 & 5 & 15 \end{array}\right]$$

Method 1 — Back-Substitution from Echelon Form

$$\text{Row 2:} \quad 5x_2 = 15 \;\Rightarrow\; x_2 = 3; \qquad \text{Row 1:} \quad x_1 - x_2 = 4 \;\Rightarrow\; x_1 - 3 = 4 \;\Rightarrow\; x_1 = 7.$$

Method 2 — Reduce Fully to RREF First

Divide Row 2 by 5 $(R_2/5 \to R_2)$:

$$\left[\begin{array}{cc|c} 1 & -1 & 4 \\ 0 & 1 & 3 \end{array}\right]$$

Add R_2 to R_1 $(R_1 + R_2 \to R_1)$:

$$\left[\begin{array}{cc|c} 1 & 0 & 7 \\ 0 & 1 & 3 \end{array}\right]$$

Unique Solution: $x_1 = 7$, $x_2 = 3$

2.11. Case 3: Infinitely Many Solutions

Condition

- The last column is **NOT** a pivot column, **AND**
- At least one free variable exists — at least one coefficient column has no pivot.

Example — RREF Matrix

$$\left[\begin{array}{ccccc|c} 1 & -1 & 0 & -5 & 0 & -9 \\ 0 & 0 & 1 & 4 & 0 & 4 \\ 0 & 0 & 0 & 0 & 1 & 3 \end{array}\right]$$

1. Identify Variables

- Pivot Columns: $1, 3, 5 \;\rightarrow\;$ Pivot Variables: x_1, x_3, x_5
- Free Columns: $2, 4 \;\rightarrow\;$ Free Variables: x_2, x_4

2. System Equations

$$\begin{aligned} x_1 - x_2 - 5x_4 &= -9 \\ x_3 + 4x_4 &= 4 \\ x_5 &= 3 \end{aligned}$$

3. General Solution (isolate the pivot variables)

$$\begin{aligned} x_1 &= x_2 + 5x_4 - 9 \\ x_2 &= \textbf{free} \\ x_3 &= -4x_4 + 4 \\ x_4 &= \textbf{free} \\ x_5 &= 3 \end{aligned}$$

Particular Solution

Setting the free variables $x_2 = 0$ and $x_4 = 0$ gives one specific solution vector:

$$(x_1, x_2, x_3, x_4, x_5) = (-9, 0, 4, 0, 3)$$

2.12. A Quick Decision Table

Table 2.1: Deciding the number of solutions from the row-reduced augmented matrix.

Last column has a pivot?	**Free variables?**	**System status**	**Number of solutions**
Yes $[0 \cdots 0 \mid \blacksquare]$	N/A	Inconsistent	0 Solutions
No	No	Consistent	1 Unique Solution
No	Yes	Consistent	Infinitely Many

2.13. Summary

- Echelon Form (REF) requires non-zero rows on top and staircase-pattern leading entries.
- RREF additionally requires all pivots to equal 1, with zeros above AND below each pivot.
- Every matrix has a unique RREF, reachable through elementary row operations.
- Pivot columns fix pivot variables; non-pivot columns give free variables.
- Reduction proceeds in two passes: forward to REF, then backward to RREF.
- A pivot in the constants column always signals an inconsistent (no-solution) system.
- No free variables and a consistent system $\rightarrow$ exactly one unique solution.
- At least one free variable and a consistent system $\rightarrow$ infinitely many solutions.

The next chapter turns from matrices to vectors: the spaces $\mathbb{R}^n$, linear combinations, and span.

Chapter 3

Vectors, Linear Combinations and Span

Part 3 · Vector Operations, Span, Existence of Solutions

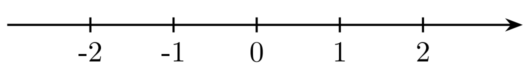
3.1. Goals

- Learn what vectors are, and what $\mathbb{R}^n$ represents.
- Define a linear combination and the span of a set of vectors.
- Understand how to determine whether a vector $\vec{b}$ can be written as a linear combination of a set of vectors — both computationally and conceptually.

3.2. The Spaces $\mathbb{R}$, $\mathbb{R}^2$, $\mathbb{R}^3$ and $\mathbb{R}^n$

$\mathbb{R}$ — Real Numbers (1-Dimensional)

Represents the real number line:



$\mathbb{R}^2$ — The Plane

$$\mathbb{R}^2 = \{ \text{ all } (x, y) \text{ such that } x, y \in \mathbb{R} \} \qquad \text{(set of ordered pairs)}$$

$\mathbb{R}^3$ — 3D Space

$$\mathbb{R}^3 = \{ \text{ all } (x, y, z) \text{ such that } x, y, z \in \mathbb{R} \} \qquad \text{(set of ordered triples)}$$

General Definition — $\mathbb{R}^n$

$$\mathbb{R}^n := \{ \text{ all possible set of ordered } n\text{-tuples } (x_1, x_2, \ldots, x_n) \text{ of reals } \}$$

where ":=" means "defined to be equal to."

Remark 3.1: *Elements of $\mathbb{R}^n$ are usually written as column vectors:*

$$\mathbb{R}^n = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \qquad \textit{where } x_1, x_2, \ldots, x_n \in \mathbb{R}.$$

3.3. Vector Notation and the Zero Vector

Definition 3.1 (General vector)**:** We write a general vector in $\mathbb{R}^n$ as

$$\vec{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n.$$

(The arrow above x represents that it is a vector.)

Definition 3.2 (Zero vector)**:**

$$\vec{0} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \in \mathbb{R}^n.$$

Remark 3.2: *The individual values inside a vector are called* entries *of the vector.*

3.4. Vector Operations

Let $\vec{x} = [x_1, \ldots, x_n]$, $\vec{y} = [y_1, \ldots, y_n] \in \mathbb{R}^n$, and let $c \in \mathbb{R}$ be a scalar (constant).

Definition 3.3 (Vector addition)**:**

$$\vec{x} + \vec{y} := \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{bmatrix}$$

Definition 3.4 (Vector subtraction)**:**

$$\vec{x} - \vec{y} := \begin{bmatrix} x_1 - y_1 \\ x_2 - y_2 \\ \vdots \\ x_n - y_n \end{bmatrix}$$

Definition 3.5 (Scalar multiplication (scaling))**:**

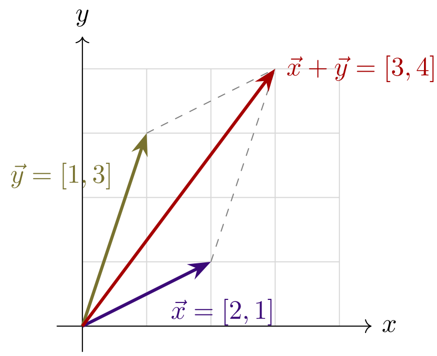
$$c\,\vec{x} = \begin{bmatrix} c \cdot x_1 \\ \vdots \\ c \cdot x_n \end{bmatrix} \qquad \text{(Scalar)} \cdot \text{(Vector)} = \text{Vector}$$

3.5. The Geometry of $\mathbb{R}^2$

Convention: always start vectors at the origin (0, 0).

Example 3.1 (Vector addition and the parallelogram rule)**:** Let $\vec{x} = [2,\ 1]$ and $\vec{y} = [1,\ 3] \in \mathbb{R}^2$:

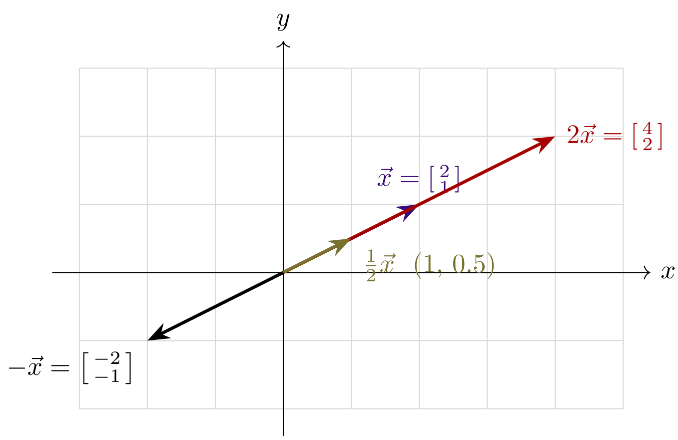
$$\vec{x} + \vec{y} = [2 + 1,\ 1 + 3] = [3,\ 4].$$



Picture in $\mathbb{R}^2$: draw $\vec{x}$, $\vec{y}$, and $\vec{x}+\vec{y}$ in the xy-plane — parallelogram addition.

Example 3.2 (Scalar scaling)**:** Let $\vec{x}=[2,\,1]$. We sketch $\vec{x}$, $2\vec{x}$, $\frac{1}{2}\vec{x}$, and $-\vec{x}$:

- $2\vec{x}=[4,\,2]$
- $\frac{1}{2}\vec{x}=[1,\,0.5]$
- $-\vec{x}=[-2,\,-1]$



3.6. Arithmetic Properties of Vectors

For all vectors $\vec{u},\vec{v},\vec{w}\in\mathbb{R}^n$ and all scalars $c,d\in\mathbb{R}$:

1. **Commutativity of addition:** $\vec{u}+\vec{v}=\vec{v}+\vec{u}$
2. **Associativity of addition:** $(\vec{u}+\vec{v})+\vec{w}=\vec{u}+(\vec{v}+\vec{w})$
3. **Additive identity:** $\vec{0}+\vec{v}=\vec{v}=\vec{v}+\vec{0}$
4. **Additive inverse:** $\vec{v}+(-\vec{v})=\vec{0}$, where $-\vec{v}=(-1)\vec{v}$
5. **Identity scalar:** $1\cdot\vec{v}=\vec{v}$
6. **Zero scalar:** $0\cdot\vec{v}=\vec{0}$
7. **Distributivity over vector addition:** $c(\vec{u}+\vec{v})=c\,\vec{u}+c\,\vec{v}$
8. **Distributivity over scalar addition:** $(c+d)\vec{v}=c\,\vec{v}+d\,\vec{v}$
9. **Associativity of scalar multiplication:** $c(d\vec{v})=(cd)\vec{v}$

3.7. Linear Combinations

Definition 3.6 (Linear combination)**:** Let $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_k$ be vectors in $\mathbb{R}^n$. Let $c_1, c_2, \ldots, c_k \in \mathbb{R}$ be scalars (called *weights*). Then the expression

$$c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_k\vec{v}_k$$

is called a **linear combination** of $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_k$.

Remark 3.3: *A linear combination of vectors results in another vector.*

3.8. Span

Definition 3.7 (Span)**:** The **span** of a set of vectors is the set of all possible linear combinations of those vectors:

$$\text{Span}\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_k\} = \text{all linear combinations of } \vec{v}_1, \ldots, \vec{v}_k.$$

Conceptually:

$$\text{Span}\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_k\} = \text{everywhere we can get to in } \mathbb{R}^n \text{ only traveling in the directions of } \vec{v}_1, \ldots, \vec{v}_k.$$

Key Fact: the zero vector $\vec{0}$ is always in $\text{Span}\{\vec{v}_1, \ldots, \vec{v}_k\}$. Why? By choosing all weights to be 0:

$$0\vec{v}_1 + 0\vec{v}_2 + \cdots + 0\vec{v}_k = \vec{0}.$$

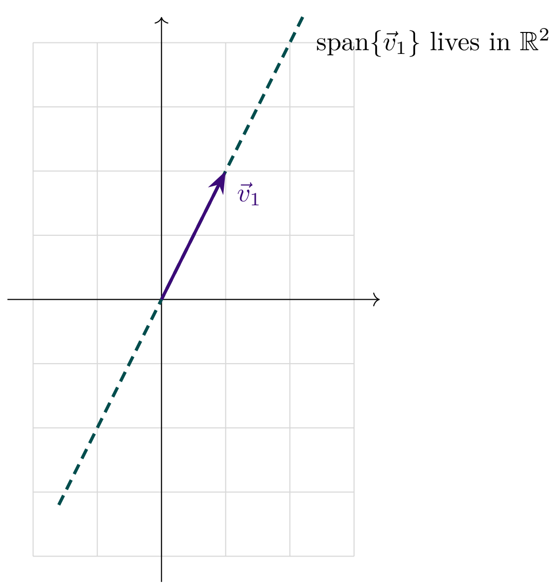
3.9. Geometric Examples of Span

(a) A Single Vector in $\mathbb{R}^2$

Example 3.3: Let $\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ in $\mathbb{R}^2$. Then

$$\text{Span}\{\vec{v}_1\} = \{\, c\,\vec{v}_1 : c \in \mathbb{R}\,\},$$

which is a line through the origin.



(b) Two Linearly Independent Vectors in $\mathbb{R}^3$

Example 3.4: Let $\vec{v}_1 = [1,\ 0,\ 0]$ and $\vec{v}_2 = [0,\ 1,\ 0]$ in $\mathbb{R}^3$:

$$c_1\vec{v}_1 + c_2\vec{v}_2 = c_1[1,0,0] + c_2[0,1,0] = [\,c_1,\ c_2,\ 0\,].$$

Geometric interpretation: The xy-plane in 3D-space ($\mathbb{R}^3$).

Remark 3.4 (Important)**:** Span$\{\vec{v}_1, \vec{v}_2\} \neq \mathbb{R}^2$, *because the vectors in this span have 3 components and live in* $\mathbb{R}^3$, *whereas elements of* $\mathbb{R}^2$ *only have 2 components.*

(c) Two Parallel (Dependent) Vectors in $\mathbb{R}^3$

Example 3.5: Let $\vec{v}_1 = [1,\ 3,\ -4]$ and $\vec{v}_2 = \left[\frac{1}{4},\ \frac{3}{4},\ -1\right]$ in $\mathbb{R}^3$. Observe that $\vec{v}_1 = 4\vec{v}_2$. Substituting into a general linear combination gives:

$$c_1\vec{v}_1 + c_2\vec{v}_2 = c_1(4\vec{v}_2) + c_2\vec{v}_2 = (4c_1 + c_2)\,\vec{v}_2.$$

Since $(4c_1 + c_2)$ is just a scalar constant:

$$\text{Span}\{\vec{v}_1,\ \vec{v}_2\} = \text{Span}\{\vec{v}_2\} = \text{the line through the origin and } \left[\tfrac{1}{4},\ \tfrac{3}{4},\ -1\right] \text{ in } \mathbb{R}^3.$$

3.10. Matrix Representation of a Linear System

Suppose a linear system has m equations and n variables. Its augmented matrix is expressed as:

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right] = \left[\begin{array}{cccc|c} \vec{a}_1 & \vec{a}_2 & \cdots & \vec{a}_n & \vec{b} \end{array}\right] = [\,A \mid \vec{b}\,]$$

- $A = [\,\vec{a}_1, \vec{a}_2, \ldots, \vec{a}_n\,]$ is the $m \times n$ **coefficient matrix**.
- Each column $\vec{a}_1, \vec{a}_2, \ldots, \vec{a}_n$ is a column vector in $\mathbb{R}^m$.
- $\vec{b} = [b_1, b_2, \ldots, b_m]$ is the right-hand side constant vector in $\mathbb{R}^m$.

3.11. Three Equivalent Descriptions of a Solution

Let $\vec{x} = [x_1, x_2, \ldots, x_n] \in \mathbb{R}^n$ be a candidate solution vector for $[\,A \mid \vec{b}\,]$. The following statements are equivalent.

1. System of Equations Representation

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

2. Vector Equation Representation

$$x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} + \cdots + x_n \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} \qquad \Longleftrightarrow \qquad x_1\vec{a}_1 + x_2\vec{a}_2 + \cdots + x_n\vec{a}_n = \vec{b}$$

3. Linear Combination / Span Interpretation

$\vec{b}$ is a linear combination of the column vectors $\vec{a}_1, \ldots, \vec{a}_n$.

Equivalently: $\vec{b} \in \mathrm{Span}\{\vec{a}_1,\ \ldots,\ \vec{a}_n\}$

3.12. The Matrix–Vector Product

Definition 3.8 (Matrix–vector product)**:** If $A = [\,\vec{a}_1\ \vec{a}_2\ \cdots\ \vec{a}_n\,]$ is an $m \times n$ matrix (with m rows and n columns, where each $\vec{a}_i \in \mathbb{R}^m$) and $\vec{x} = [x_1, x_2, \ldots, x_n] \in \mathbb{R}^n$, then the product $A\vec{x}$ is defined as:

$$A\vec{x} := x_1\vec{a}_1 + x_2\vec{a}_2 + \cdots + x_n\vec{a}_n$$

Remark 3.5: *This product $A\vec{x}$ represents a linear combination of the columns of A with weights given by the entries of $\vec{x}$, resulting in a vector in $\mathbb{R}^m$.*

3.13. Existence of Solutions

Theorem 3.1 (Existence of solutions)**:** *The following statements are equivalent:*

The linear system $[\,A \mid \vec{b}\,]$ has a solution

- $\Longleftrightarrow$ *$\vec{b}$ is a linear combination of $\vec{a}_1, \ldots, \vec{a}_n$ (same as saying $\vec{b} \in \mathrm{span}\{\vec{a}_1, \ldots, \vec{a}_n\}$);*
- $\Longleftrightarrow$ *$A\vec{x} = \vec{b}$ for some $\vec{x} \in \mathbb{R}^n$;*
- $\Longleftrightarrow$ *the echelon form of $[\,A \mid \vec{b}\,]$ has NO row of the form $[\,0\ \cdots\ 0 \mid \blacksquare\,]$ with $\blacksquare \neq 0$.*

3.14. Worked Example: Testing Span Membership

Example 3.6: Problem: is

$$[1,\ 2,\ 3] \in \mathrm{Span}\{\,[-1, 1, 2],\ [2, 1, 0],\ [1, 2, 2]\,\}\,?$$

(i.e., is $[1, 2, 3]$ a linear combination of those vectors?)

Solution: Check whether the linear system given by the augmented matrix is consistent. Starting matrix:

$$\left[\begin{array}{ccc|c} -1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 2 \\ 2 & 0 & 2 & 3 \end{array}\right]$$

Apply row operations $R_1 + R_2 \to R_2,\ 2R_1 + R_3 \to R_3$:

$$\left[\begin{array}{ccc|c} -1 & 2 & 1 & 1 \\ 0 & 3 & 3 & 3 \\ 0 & 4 & 4 & 5 \end{array}\right]$$

Simplify rows: $(-1)R_1 \to R_1,\ (1/3)R_2 \to R_2$:

$$\left[\begin{array}{ccc|c} 1 & -2 & -1 & -1 \\ 0 & 1 & 1 & 1 \\ 0 & 4 & 4 & 5 \end{array}\right]$$

Eliminate Row 3: $-4R_2 + R_3 \to R_3$:

$$\left[\begin{array}{ccc|c} 1 & -2 & -1 & -1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{array}\right]$$

The row $[\,0\ 0\ 0 \mid 1\,]$ indicates that the system is inconsistent.

$[1,\ 2,\ 3]$ **is NOT in** $\mathrm{Span}\{\,[-1,1,2],\ [2,1,0],\ [1,2,2]\,\}$

3.15. Worked Example: Finding Weights in a Span

Example 3.7: Problem: is

$$\vec{b} = [-9,\ 3,\ 10] \in \mathrm{Span}\{\,[-1,1,2],\ [2,1,0]\,\}\,?$$

(If yes, find weights x_1, x_2 such that $\vec{b} = x_1\vec{v}_1 + x_2\vec{v}_2$.)

Solution: Set up the augmented matrix:

$$\left[\begin{array}{cc|c} -1 & 2 & -9 \\ 1 & 1 & 3 \\ 2 & 0 & 10 \end{array}\right]$$

Row operations to Echelon Form: $R_1 + R_2 \to R_2,\ 2R_1 + R_3 \to R_3$:

$$\left[\begin{array}{cc|c} -1 & 2 & -9 \\ 0 & 3 & -6 \\ 0 & 4 & -8 \end{array}\right]$$

Rescale rows: $R_1(-1) \to R_1,\ R_2(1/3) \to R_2,\ R_3(1/4) \to R_3$:

$$\left[\begin{array}{cc|c} 1 & -2 & 9 \\ 0 & 1 & -2 \\ 0 & 1 & -2 \end{array}\right]$$

Eliminate Row 3: $R_2 - R_3 \to R_3$, giving the Echelon Form:

$$\left[\begin{array}{cc|c} 1 & -2 & 9 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{array}\right]$$

Observation: the Echelon Form has no row of the form $[\,0\ldots0 \mid \blacksquare\,]$ where $\blacksquare \neq 0$ — thus a solution exists.

Back substitution / RREF: $2R_2 + R_1 \to R_1$:

$$\left[\begin{array}{cc|c} 1 & 0 & 5 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{array}\right] \qquad \Longrightarrow \qquad x_1 = 5, \quad x_2 = -2.$$

Thus:

$$[-9,\, 3,\, 10] \;=\; 5 \cdot [-1,\, 1,\, 2] \;-\; 2 \cdot [2,\, 1,\, 0]$$

3.16. Spanning the Entire Space

Theorem 3.2 (Spanning $\mathbb{R}^m$)**:** *Let A be an $m \times n$ matrix with columns $\vec{a}_1, \ldots, \vec{a}_n$. The following statements are equivalent ($\Longleftrightarrow$):*

1. *The linear system $[\,A \mid \vec{b}\,]$ has a solution for every $\vec{b} \in \mathbb{R}^m$.*
2. $\text{Span}\{\vec{a}_1, \ldots, \vec{a}_n\} = \mathbb{R}^m$.
3. *The reduced matrix never has a row of the form $[\,0\ldots0 \mid \blacksquare\,]$ (where $\blacksquare \neq 0$).*
4. *The coefficient matrix A has a pivot in every row.*

3.17. Worked Example: Testing a Full-Space Span

Example 3.8: Problem: Do the vectors $[1, 2, 3]$, $[1, 0, 1]$, $[2, -1, 1]$ span all of $\mathbb{R}^3$?

Solution: Form the matrix A and check whether there is a pivot in each row:

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 0 & -1 \\ 3 & 1 & 1 \end{bmatrix}$$

Eliminate the lower entries in Column 1: $-2R_1 + R_2 \to R_2$, $-3R_1 + R_3 \to R_3$:

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & -2 & -5 \\ 0 & -2 & -5 \end{bmatrix}$$

Eliminate Row 3: $R_3 - R_2 \to R_3$:

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & -2 & -5 \\ 0 & 0 & 0 \end{bmatrix}$$

Since row 3 contains all zeros, not every row has a pivot (there are only 2 pivots, but 3 rows are required).

The columns do NOT span $\mathbb{R}^3$.

3.18. Summary

- $\mathbb{R}^n$ is the set of ordered n-tuples, usually written as column vectors.
- Vectors can be added, subtracted, and scaled entrywise.
- Vector addition follows the parallelogram rule geometrically.
- A linear combination $c_1\vec{v}_1 + \cdots + c_k\vec{v}_k$ always produces another vector.
- $\text{Span}\{\vec{v}_1, \ldots, \vec{v}_k\}$ is the set of ALL linear combinations — and always contains $\vec{0}$.
- A linear system $[\,A \mid \vec{b}\,]$ has a solution $\iff \vec{b} \in \text{Span}\{\text{columns of } A\} \iff A\vec{x} = \vec{b}$ for some $\vec{x}$.
- A pivot-free row in the constants column of the echelon form guarantees a solution exists.
- Columns of A span the ENTIRE space $\mathbb{R}^m$ only when A has a pivot in every row.

The next chapter studies the matrix equation $A\vec{x} = \vec{b}$ in its own right.

Chapter 4

The Matrix Equation $A\vec{x} = \vec{b}$

Part 4 · Matrix–Vector Multiplication, Vector and Matrix Form

4.1. Goals

- **Goal 1:** learn how to compute $A\vec{x}$ in two ways.
- **Goal 2:** understand how to write a linear system in **vector form** and **matrix form**.

4.2. Defining Matrix–Vector Multiplication

Definition 4.1 (Matrix–vector multiplication)**:** Let $A = [\,\vec{a}_1\ \vec{a}_2\ \cdots\ \vec{a}_n\,]$ be an $m \times n$ matrix (m rows, n columns), where each column vector $\vec{a}_i \in \mathbb{R}^m$. Let

$$\vec{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n.$$

Then the product $A\vec{x}$ is defined as a **linear combination of the columns of** A:

$$A\vec{x} \;=\; \begin{bmatrix} \vec{a}_1 & \vec{a}_2 & \cdots & \vec{a}_n \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \;:=\; x_1\vec{a}_1 + x_2\vec{a}_2 + \cdots + x_n\vec{a}_n$$

4.3. Computing $A\vec{x}$: Two Methods

Example 4.1: Compute the following, if defined.

Part (a)

$$\begin{bmatrix} 0 & -3 & 5 \\ 2 & 1 & 4 \end{bmatrix} \begin{bmatrix} 5 \\ -1 \\ 2 \end{bmatrix}$$

Method 1: Linear Combination of Columns.

$$= 5\begin{bmatrix} 0 \\ 2 \end{bmatrix} - 1\begin{bmatrix} -3 \\ 1 \end{bmatrix} + 2\begin{bmatrix} 5 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 10 \end{bmatrix} + \begin{bmatrix} 3 \\ -1 \end{bmatrix} + \begin{bmatrix} 10 \\ 8 \end{bmatrix} = \begin{bmatrix} \mathbf{13} \\ \mathbf{17} \end{bmatrix}$$

Method 2: Row-Vector Rule (Dot Product Rule).

$$\begin{bmatrix} 5(0) + (-1)(-3) + 2(5) \\ 5(2) + (-1)(1) + 2(4) \end{bmatrix} = \begin{bmatrix} \mathbf{13} \\ \mathbf{17} \end{bmatrix}$$

Part (b)

$$\begin{bmatrix} 0 & -3 & 5 \\ 2 & 1 & 4 \end{bmatrix} \begin{bmatrix} 5 \\ -1 \end{bmatrix}$$

- **Result:** Undefined.
- **Reason:** the number of columns of A (3) does not match the number of entries in the vector $\vec{x}$ (2).

Part (c)

$$\begin{bmatrix} 1 & -2 & 1 \\ 0 & 0 & 5 \\ 0 & 3 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ 0 \\ 7 \end{bmatrix}$$

Using the Row-Vector Rule:

$$= \begin{bmatrix} 3(1) + 0(-2) + 7(1) \\ 3(0) + 0(0) + 7(5) \\ 3(0) + 0(3) + 7(-1) \end{bmatrix} = \begin{bmatrix} \mathbf{10} \\ \mathbf{35} \\ \mathbf{-7} \end{bmatrix}$$

4.4. Vector Form and Matrix Form of a System

Example 4.2: Given the system of linear equations:

$$\begin{aligned} 6x_1 - x_2 &= 9 \\ 4x_1 + 3x_2 &= -5 \\ x_1 + 7x_2 &= -20 \end{aligned}$$

Vector form:

$$x_1 \begin{bmatrix} 6 \\ 4 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} -1 \\ 3 \\ 7 \end{bmatrix} = \begin{bmatrix} 9 \\ -5 \\ -20 \end{bmatrix}$$

Matrix form $(A\vec{x} = \vec{b})$:

$$\begin{bmatrix} 6 & -1 \\ 4 & 3 \\ 1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 9 \\ -5 \\ -20 \end{bmatrix}$$

4.5. Solving the Matrix Equation

Example 4.3 (continuation of Example 4.2)**:** Solve $A\vec{x} = \vec{b}$ using row reduction.

1. **Augmented matrix:**

$$\left[\begin{array}{cc|c} 6 & -1 & 9 \\ 4 & 3 & -5 \\ 1 & 7 & -20 \end{array}\right]$$

2. **Swap rows** $(R_1 \leftrightarrow R_3)$:

$$\Rightarrow \left[\begin{array}{cc|c} 1 & 7 & -20 \\ 4 & 3 & -5 \\ 6 & -1 & 9 \end{array}\right]$$

3. **Eliminate lower column 1 entries:** $-4R_1 + R_2 \to R_2$; $-6R_1 + R_3 \to R_3$:

$$\Rightarrow \left[\begin{array}{cc|c} 1 & 7 & -20 \\ 0 & -25 & 75 \\ 0 & -43 & 129 \end{array}\right]$$

4. **Rescale rows:** $R_2\left(-\frac{1}{25}\right) \to R_2$; $R_3\left(-\frac{1}{43}\right) \to R_3$:

$$\Rightarrow \left[\begin{array}{cc|c} 1 & 7 & -20 \\ 0 & 1 & -3 \\ 0 & 1 & -3 \end{array}\right]$$

5. **Eliminate Row 3:** $R_3 - R_2 \to R_3$:

$$\Rightarrow \left[\begin{array}{cc|c} 1 & 7 & -20 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{array}\right]$$

6. **Reduced Row Echelon Form:** $-7R_2 + R_1 \to R_1$:

$$\Rightarrow \left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{array}\right]$$

Final Solution: $x_1 = 1$, $x_2 = -3$, **i.e.** $\vec{x} = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$

The next chapter examines the structure of solution sets: homogeneous and nonhomogeneous systems, and the geometry behind them.

Chapter 5

Solution Sets of Linear Systems

Part 5 · Homogeneous & Nonhomogeneous Systems, Parametric Vector Form

5.1. Goals

- **Goal 1:** learn how to write the general solution to a linear system in **parametric vector form**.
- **Goal 2:** understand geometrically the difference in the solution sets of $A\vec{x} = \vec{0}$ and $A\vec{x} = \vec{b}$ when $\vec{b} \neq \vec{0}$.

5.2. Homogeneous Linear Systems

Definition 5.1 (Homogeneous system)**:** A linear system is called **homogeneous** if it can be written in the form

$$A\vec{x} = \vec{0},$$

where A is an $m \times n$ matrix and $\vec{0}$ is the zero vector in $\mathbb{R}^m$.

In terms of the columns of A:

$$A\vec{x} = \vec{0} \iff x_1\vec{a}_1 + x_2\vec{a}_2 + \cdots + x_n\vec{a}_n = \vec{0}.$$

Remark 5.1: *The vector $\vec{x} = \vec{0}$ (i.e., $x_1 = x_2 = \cdots = x_n = 0$) is **always** a solution to a homogeneous system. This is called the **trivial solution**. Nonzero solutions, if they exist, are called **nontrivial solutions**.*

5.3. Worked Example: A Homogeneous System

Example 5.1: Find the solution set of the following system in **parametric vector form**:

$$\begin{aligned} x_3 + 4x_4 - x_5 &= 0 \\ 2x_1 - 2x_2 + 4x_3 + 6x_4 + 3x_5 &= 0 \\ x_1 - x_2 + x_3 - x_4 + 2x_5 &= 0 \end{aligned}$$

Solution:

1. **Augmented matrix & row reduction to RREF:**

$$\left[\begin{array}{ccccc|c} 0 & 0 & 1 & 4 & -1 & 0 \\ 2 & -2 & 4 & 6 & 3 & 0 \\ 1 & -1 & 1 & -1 & 2 & 0 \end{array}\right] \sim \left[\begin{array}{ccccc|c} 1 & -1 & 0 & -5 & 0 & 0 \\ 0 & 0 & 1 & 4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{array}\right]$$

2. **Identify variables:** pivot variables x_1, x_3, x_5; free variables x_2, x_4.

3. **Express pivot variables in terms of the free variables:**

$$\begin{aligned} x_1 - x_2 - 5x_4 = 0 &\implies x_1 = x_2 + 5x_4 \\ x_3 + 4x_4 = 0 &\implies x_3 = -4x_4 \\ x_5 = 0 &\implies x_5 = 0 \end{aligned}$$

Writing every variable explicitly:

$$\begin{array}{lll} x_1 = x_2 + 5x_4 & x_2 = \textit{free} & x_3 = -4x_4 \\ x_4 = \textit{free} & x_5 = 0 & \end{array}$$

4. **General solution in parametric vector form:**

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} x_2 + 5x_4 \\ x_2 \\ -4x_4 \\ x_4 \\ 0 \end{bmatrix} = x_2 \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 5 \\ 0 \\ -4 \\ 1 \\ 0 \end{bmatrix}$$

Parametric vector form: $\vec{x} = x_2 \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 5 \\ 0 \\ -4 \\ 1 \\ 0 \end{bmatrix}$

5.4. Nonhomogeneous Linear Systems

Definition 5.2 (Nonhomogeneous system)**:** A linear system is called **nonhomogeneous** if it can be written in the form

$$A\vec{x} = \vec{b} \quad \text{where} \quad \vec{b} \neq \vec{0}.$$

5.5. Worked Example: A Nonhomogeneous System

Example 5.2: Find the solution set of the following system in **parametric vector form**:

$$\begin{aligned} x_3 + 4x_4 - x_5 &= 1 \\ 2x_1 - 2x_2 + 4x_3 + 6x_4 + 3x_5 &= 7 \\ x_1 - x_2 + x_3 - x_4 + 2x_5 &= 1 \end{aligned}$$

Solution:

1. **Augmented matrix & row reduction to RREF:**

$$\left[\begin{array}{ccccc|c} 0 & 0 & 1 & 4 & -1 & 1 \\ 2 & -2 & 4 & 6 & 3 & 7 \\ 1 & -1 & 1 & -1 & 2 & 1 \end{array}\right] \sim \left[\begin{array}{ccccc|c} 1 & -1 & 0 & -5 & 0 & -9 \\ 0 & 0 & 1 & 4 & 0 & 4 \\ 0 & 0 & 0 & 0 & 1 & 3 \end{array}\right]$$

2. **Express pivot variables in terms of the free variables:**

$$\begin{aligned} x_1 - x_2 - 5x_4 = -9 &\implies x_1 = -9 + x_2 + 5x_4 \\ x_3 + 4x_4 = 4 &\implies x_3 = 4 - 4x_4 \\ x_5 = 3 &\implies x_5 = 3 \end{aligned}$$

Writing every variable explicitly:

$$\begin{array}{lll} x_1 = -9 + x_2 + 5x_4 & x_2 = \textit{free} & x_3 = 4 - 4x_4 \\ x_4 = \textit{free} & x_5 = 3 & \end{array}$$

3. **General solution in parametric vector form:**

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -9 + x_2 + 5x_4 \\ x_2 \\ 4 - 4x_4 \\ x_4 \\ 3 \end{bmatrix} = \begin{bmatrix} -9 \\ 0 \\ 4 \\ 0 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 5 \\ 0 \\ -4 \\ 1 \\ 0 \end{bmatrix}$$

Parametric vector form: $\vec{x} = \begin{bmatrix} -9 \\ 0 \\ 4 \\ 0 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 5 \\ 0 \\ -4 \\ 1 \\ 0 \end{bmatrix}$

5.6. The Solution Structure Theorem

Notice how the solution vector $\vec{x}$ to $A\vec{x} = \vec{b}$ in Example 5.2 splits into two distinct components:

$$\vec{x} = \underbrace{\begin{bmatrix} -9 \\ 0 \\ 4 \\ 0 \\ 3 \end{bmatrix}}_{\vec{v}_p} + \underbrace{x_2 \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 5 \\ 0 \\ -4 \\ 1 \\ 0 \end{bmatrix}}_{\vec{v}_h}$$

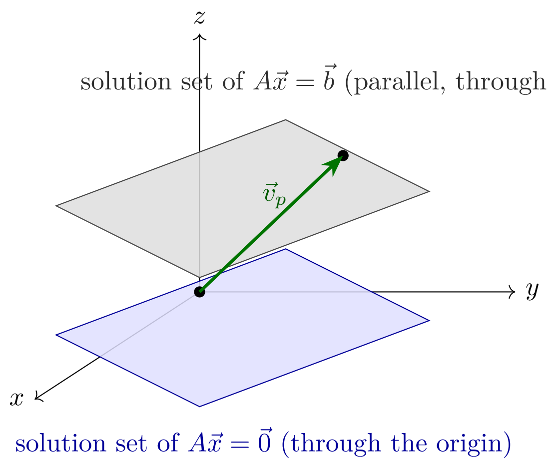
- $\vec{v}_p$: a **particular solution** to $A\vec{x} = \vec{b}$ (satisfies $A\vec{v}_p = \vec{b}$).
- $\vec{v}_h$: the **homogeneous solution** (the general solution to $A\vec{x} = \vec{0}$).

Theorem 5.1 (Solution structure)**:** *Suppose the equation $A\vec{x} = \vec{b}$ is consistent for a given vector $\vec{b}$, and let $\vec{v}_p$ be a particular solution. Then the solution set of $A\vec{x} = \vec{b}$ (where $\vec{b} \neq \vec{0}$) is the set of all vectors of the form*

$$\vec{x} = \vec{v}_p + \vec{v}_h,$$

where $\vec{v}_h$ is any solution of the corresponding homogeneous system $A\vec{x} = \vec{0}$.

5.7. The Geometric Picture in $\mathbb{R}^3$



- The solution to $A\vec{x} = \vec{b}$ is **parallel** to the solution to $A\vec{x} = \vec{0}$ and goes through $\vec{v}_p$.
- The solution to $A\vec{x} = \vec{0}$ goes through the origin, since $\vec{x} = \vec{0}$ is a solution to $A\vec{x} = \vec{0}$.

The geometric picture in $\mathbb{R}^3$ for a consistent linear system $A\vec{x} = \vec{b}$ is an **affine plane**: a translation of the solution subspace of the homogeneous system $A\vec{x} = \vec{0}$ (a plane through the origin) by any particular solution $\vec{v}_p$ of $A\vec{x} = \vec{b}$.

Remark 5.2: *If $A\vec{x} = \vec{b}$ is inconsistent, there is no geometric picture (no intersection). If it has a unique solution, the picture reduces to a single point.*

Summary of the picture: the solution set of $A\vec{x} = \vec{b}$ is **parallel** to the solution set of $A\vec{x} = \vec{0}$, and passes through $\vec{v}_p$. The solution set of $A\vec{x} = \vec{0}$ passes through the origin, since $\vec{x} = \vec{0}$ is always a solution to $A\vec{x} = \vec{0}$.

The second half of the book begins: linear independence.

Chapter 6

Linear Independence

Part 6 · Concepts, Tests, and Conceptual Reasoning

6.1. Overview and Goals

- **Goal 1:** learn what **linearly independent** and **linearly dependent** vectors are conceptually, and be able to check independence computationally.
- **Goal 2:** use the concept of independence to tackle conceptual examples.

6.2. What Is Linear Independence?

Definition 6.1 (Linear independence)**:** Let $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_k \in \mathbb{R}^n$. The set $\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_k\}$ is **linearly independent** if the vector equation

$$c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_k\vec{v}_k = \vec{0}$$

only has the **trivial solution**:

$$c_1 = c_2 = \cdots = c_k = 0.$$

In matrix form, $A\vec{x} = \vec{0}$ (where $A = [\,\vec{v}_1\ \vec{v}_2\ \ldots\ \vec{v}_k\,]$ and

$$\vec{x} = \begin{bmatrix} c_1 \\ \vdots \\ c_k \end{bmatrix}$$

) only has the solution $\vec{x} = \vec{0}$.

Definition 6.2 (Linear dependence)**:** If there exists a solution where **not all** $c_i = 0$, we say the set $\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_k\}$ is **linearly dependent**.

6.3. Example 1: Demonstrating Linear Dependence

Example 6.1: Given the combination:

$$4\begin{bmatrix} 5 \\ 3 \\ 1 \end{bmatrix} - 10\begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} + 2\begin{bmatrix} 0 \\ -1 \\ 13 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Because there exists a non-trivial solution ($c_1 = 4,\ c_2 = -10,\ c_3 = 2$), the set of vectors

$$\left\{ \begin{bmatrix} 5 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 13 \end{bmatrix} \right\}$$

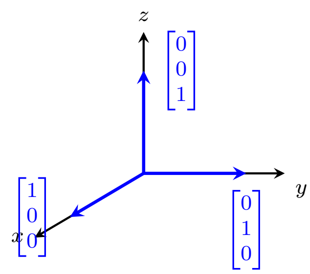
is linearly dependent.

6.4. Intuitive and Geometric Interpretation

Intuitively, $\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_k\}$ is linearly independent if the vectors point in **"completely independent"** directions.

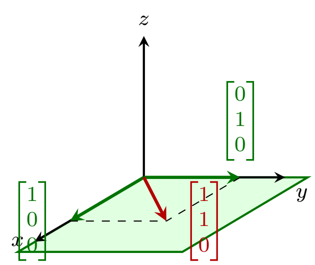
Comparison in $\mathbb{R}^3$

Linearly independent.



The set $\left\{\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}\right\}$ is **linearly independent**: the three vectors are not in the same plane, and no vector can be written as a linear combination of the other two.

Linearly dependent.



The set $\left\{\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}1\\1\\0\end{bmatrix}\right\}$ is **linearly dependent**: the green shaded plane is $\operatorname{span}\{\vec{e}_1, \vec{e}_2\}$ (the xy-plane), and $\begin{bmatrix}1\\1\\0\end{bmatrix}$ lies inside that span, since $\begin{bmatrix}1\\1\\0\end{bmatrix} = \begin{bmatrix}1\\0\\0\end{bmatrix} + \begin{bmatrix}0\\1\\0\end{bmatrix}$ (dashed lines show the parallelogram rule).

6.5. Key Theorem and Caution

Theorem 6.1 (Characterisation of dependence)**:** *A set of two or more vectors $\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_k\}$ is **linearly dependent** if and only if **at least one** vector $\vec{v}_i$ is a linear combination of the other vectors.*

Corollary: Any set of vectors containing the zero vector $\vec{0}$ is automatically linearly dependent. (A one-element set $\{\vec{v}\}$ is dependent exactly when $\vec{v} = \vec{0}$.)

Remark 6.1 (Caution)**:** *It is* ***not necessarily true*** *that* every *vector $\vec{v}_i$ in a dependent set is a linear combination of the other vectors.*

6.6. Testing for Linear Independence

Key Criterion:

Theorem 6.2 (Pivot test)**:** *The vectors $\vec{a}_1, \vec{a}_2, \ldots, \vec{a}_k$ are* ***linearly independent*** *if and only if $A\vec{x} = \vec{0}$ has a* ***unique solution*** *(i.e., $\vec{x} = \vec{0}$).*

$$\Updownarrow$$

Matrix A has ***no free variables*** *(i.e., there is a* ***pivot in every column*** *of A).*

6.7. Example 2: Testing Sets for Independence

Example 6.2: Problem: test if the following sets of vectors are linearly independent. If not, express a linear dependence relationship among them.

Part (a)

$$\left\{ \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} \right\}$$

Solution: form the matrix A using the vectors as columns and reduce to row-echelon form:

$$A = \begin{bmatrix} -1 & 2 & 1 \\ 1 & 1 & 2 \\ 2 & 0 & 4 \end{bmatrix}$$

1. Performing $(R_1 + R_2 \to R_2$ and $2R_1 + R_3 \to R_3)$:

$$\begin{bmatrix} -1 & 2 & 1 \\ 0 & 3 & 3 \\ 0 & 4 & 6 \end{bmatrix}$$

2. Performing $\left(\frac{1}{3}R_2 \to R_2\right)$:

$$\begin{bmatrix} -1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 4 & 6 \end{bmatrix}$$

3. Performing $(-4R_2 + R_3 \to R_3)$:

$$\begin{bmatrix} -1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

- **Conclusion:** since there is a **pivot in every column**, there are no free variables. Therefore, the column vectors are **linearly independent**.

The set in Part (a) is linearly independent.

Part (b)

$$\left\{ \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 12 \\ 3 \\ -4 \end{bmatrix} \right\}$$

Solution: Form the matrix A and row-reduce to reduced row echelon form (RREF):

$$\begin{bmatrix} -1 & 2 & 12 \\ 1 & 1 & 3 \\ 2 & 0 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 5 \\ 0 & 0 & 0 \end{bmatrix}$$

- **Analysis:** not all columns have pivots (the 3rd column lacks a pivot), so x_3 is a free variable. Thus, the column vectors are **linearly dependent**.

Finding the Linear Dependence Relationship

Set up the homogeneous system $A\vec{x} = \vec{0}$:

$$\begin{aligned} x_1 - 2x_3 = 0 \quad &\implies \quad x_1 = 2x_3 \\ x_2 + 5x_3 = 0 \quad &\implies \quad x_2 = -5x_3 \\ x_3 &= \textit{free} \end{aligned}$$

Writing the solution in parametric vector form:

$$\vec{x} = x_3 \begin{bmatrix} 2 \\ -5 \\ 1 \end{bmatrix}$$

Choosing $x_3 = 1$ gives $x_1 = 2$, $x_2 = -5$, and $x_3 = 1$.

Linear Dependence Relationship

$$2 \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} - 5 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} 12 \\ 3 \\ -4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

The set in Part (b) is linearly dependent, with $2\vec{v}_1 - 5\vec{v}_2 + \vec{v}_3 = \vec{0}$.

6.8. Conceptual Examples

Example 3

Example 6.3: Problem: can a set of 4 vectors in $\mathbb{R}^3$ be linearly independent? If yes, give an example. If no, explain why.

Solution: Answer: No.

Proof / Explanation: let $\vec{v}_1, \vec{v}_2, \vec{v}_3, \vec{v}_4 \in \mathbb{R}^3$. Form the matrix A using these vectors as columns:

$$A = \begin{bmatrix} \vec{v}_1 & \vec{v}_2 & \vec{v}_3 & \vec{v}_4 \end{bmatrix}$$

- **Matrix Dimensions:** since each vector belongs to $\mathbb{R}^3$, matrix A has 3 rows and 4 columns (3×4).
- **Pivot Constraint:** there are only 3 rows, so the matrix can have at most 3 pivots.
- **Column Analysis:** since A has 4 columns and at most 3 pivots, **not all columns can contain a pivot** (there is at least one free variable).
- **Conclusion:** since not every column has a pivot, the column vectors are **linearly dependent**.

Remark 6.2 (Intuition)**:** *There cannot be 4 independent directions in a 3-dimensional space ($\mathbb{R}^3$).*

General Theorem: if a matrix A is $m \times n$ with $n > m$ (i.e., there are more vectors than elements in each vector), then its columns are **always linearly dependent**.

Example 4

Example 6.4 (True / False)**:** If $\vec{v}_1, \vec{v}_2, \vec{v}_3 \in \mathbb{R}^3$ and $\vec{v}_1$ is **not** a linear combination of $\vec{v}_2$ and $\vec{v}_3$, then $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ is linearly independent. *(If true, explain why. If false, give a counterexample.)*

Key Theorem (Theorem 6.1): a set of vectors $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k\}$ is **linearly dependent** $\iff$ **at least one** vector $\vec{v}_i$ is a linear combination of the other vectors.

(Note: we don't necessarily know which *specific vector it is.)*

Solution: Answer: False.

Counterexample: choose the following vectors in $\mathbb{R}^3$:

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \qquad \vec{v}_2 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \qquad \vec{v}_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

1. **Verify the hypothesis** ($\vec{v}_1$ is not a linear combination of $\vec{v}_2$ and $\vec{v}_3$). Any linear combination of $\vec{v}_2$ and $\vec{v}_3$ gives:
$$a \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \neq \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$
Therefore, $\vec{v}_1$ is **not** a linear combination of $\vec{v}_2$ and $\vec{v}_3$.

2. **Test for linear independence of the set $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$.** Form the matrix $A = [\,\vec{v}_1\ \vec{v}_2\ \vec{v}_3\,]$:
$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$
   - Matrix A has only 1 pivot (in the 1^{st} column).
   - It **doesn't** have pivots in each column (Columns 2 and 3 lack pivots).

- Therefore, the set $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ is **linearly dependent**.

The statement is False: the set is linearly dependent (it contains the zero vector).

END OF PART 6 · NEXT: PART 7 — LINEAR TRANSFORMATIONS

Chapter 7

Linear Transformations

Part 7 · Sections 1.8–1.9 · Standard Matrices, Onto & One-to-One Mappings

7.1. Overview and Goals

1. **Define linear transformation** and understand the connection between linear transformations and matrices.
2. **Learn how to construct** the **standard matrix** of a linear transformation.
3. **Be able to determine** if a linear transformation T is:
   - **(1) Onto**
   - **(2) One-to-one**

7.2. Matrices as Transformations (Functions)

Let A be an $m \times n$ matrix (with m rows and n columns) and

$$\vec{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n.$$

We can think of matrix multiplication as a **function (transformation)**:

- **Input:** $\vec{x} \in \mathbb{R}^n$
- **Output:** $A\vec{x} = x_1\vec{a}_1 + \cdots + x_n\vec{a}_n \in \mathbb{R}^m$
- **Function Mapping:** $T_A : \mathbb{R}^n \longrightarrow \mathbb{R}^m$

Definition 7.1 (Linear transformation associated with a matrix)**:** Given A, an $m \times n$ matrix, we can construct a function $T_A : \mathbb{R}^n \to \mathbb{R}^m$. We call T_A the **linear transformation** associated with A:

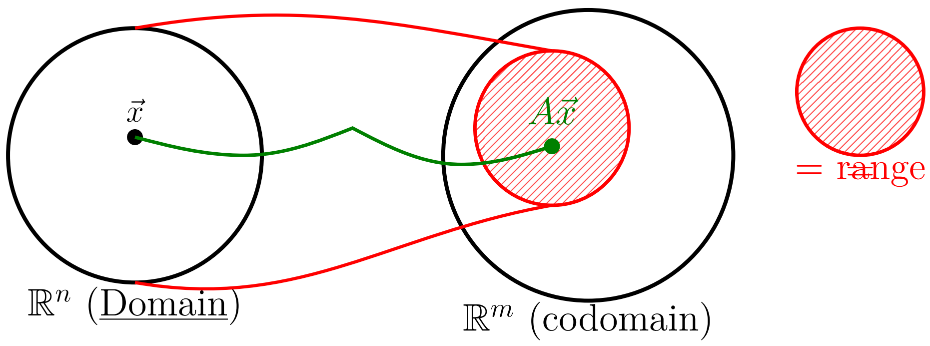
$$\begin{aligned} T_A : \mathbb{R}^n &\longrightarrow \mathbb{R}^m \qquad (\underline{\text{Domain}} \to \text{Codomain}) \\ \vec{x} &\longmapsto A\vec{x} \qquad (\text{Input} \to \text{Output}) \end{aligned}$$

Key Terminology

- **Domain:** $\mathbb{R}^n$ (the set of all valid inputs)
- **Codomain:** $\mathbb{R}^m$ (all potential outputs)
- **Range:** all *achieved* outputs

$$\{\vec{b} \in \mathbb{R}^m \mid \vec{b} = A\vec{x} \text{ for some } \vec{x} \in \mathbb{R}^n\}$$

Note: Codomain and Range are not always equal.



7.3. Example 1: Computing a Transformation

Example 7.1: Let

$$A = \begin{bmatrix} 1 & 4 \\ 2 & 8 \\ 3 & 12 \end{bmatrix}.$$

Since A is 3×2, matrix multiplication requires $A_{(3\times2)}\vec{x}_{(2\times1)}$. Therefore, the mapping is

$$T_A : \mathbb{R}^2 \longrightarrow \mathbb{R}^3 \qquad \text{(Domain} \to \text{Codomain).}$$

$$T_A\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = x_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} 4 \\ 8 \\ 12 \end{bmatrix}$$

- **Specific Evaluation:**

$$T_A\left(\begin{bmatrix} 6 \\ -1 \end{bmatrix}\right) = 6 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} - 1 \begin{bmatrix} 4 \\ 8 \\ 12 \end{bmatrix} = \begin{bmatrix} \mathbf{2} \\ \mathbf{4} \\ \mathbf{6} \end{bmatrix}$$

  We say

$$\begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}$$

is the **image** of

$$\begin{bmatrix} 6 \\ -1 \end{bmatrix} \qquad \text{under } T_A.$$

Analyzing Range vs. Codomain

Notice the columns of A are scalar multiples:

$$x_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + 4x_2 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = (x_1 + 4x_2) \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

(where $(x_1 + 4x_2)$ is an arbitrary constant)

- Range of A = a line in $\mathbb{R}^3$
- Codomain of $A = \mathbb{R}^3$
- $\implies$ Range of $A \neq$ Codomain of A

Existence Condition:

$$\vec{b} \in \text{Range}(A) \iff A\vec{x} = \vec{b} \text{ is consistent}$$

7.4. Properties of a Linear Transformation

1. **Additivity:**
$$A(\vec{u} + \vec{v}) = A\vec{u} + A\vec{v} \quad \implies \quad T_A(\vec{u} + \vec{v}) = T_A(\vec{u}) + T_A(\vec{v})$$

2. **Scalar Multiplication:**
$$A(c\vec{u}) = c(A\vec{u}) \quad \implies \quad T_A(c\vec{u}) = c\,T_A(\vec{u})$$

Core Insight: $T_A : \mathbb{R}^n \to \mathbb{R}^m$ preserves addition and scalar multiplication of vectors.

This property is rare! Most arbitrary functions do not preserve these operations.

7.5. Non-Example: A Function Failing Linearity

Consider $f : \mathbb{R}^2 \to \mathbb{R}^2$ defined by:

$$f\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1^2 + 1 \\ x_1 + x_2 \end{bmatrix}$$

Testing additivity:

- **Evaluating sum first:**
$$f\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \end{bmatrix}\right) = f\left(\begin{bmatrix} 3 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 3^2 + 1 \\ 3 + 2 \end{bmatrix} = \begin{bmatrix} 10 \\ 5 \end{bmatrix}$$

- **Evaluating terms separately:**
$$f\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) + f\left(\begin{bmatrix} 2 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1^2 + 1 \\ 1 + 1 \end{bmatrix} + \begin{bmatrix} 2^2 + 1 \\ 2 + 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix} + \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 7 \\ 5 \end{bmatrix}$$

Since

$$\begin{bmatrix} 10 \\ 5 \end{bmatrix} \neq \begin{bmatrix} 7 \\ 5 \end{bmatrix}, \qquad f \text{ is } \textbf{not} \text{ linear.}$$

7.6. Formal Definition of Linear Transformation

Definition 7.2 (Linear transformation)**:** A function $T : \mathbb{R}^n \to \mathbb{R}^m$ is a **linear transformation** if, for all $\vec{u}, \vec{v} \in \mathbb{R}^n$ and scalars c:

1. $T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$
2. $T(c\vec{u}) = c\,T(\vec{u})$

Theorem: if A is an $m \times n$ matrix, then $T_A : \mathbb{R}^n \to \mathbb{R}^m$ is **always** a linear transformation.

7.7. Standard Matrix for Linear Transformations

Core Problem

Given a linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$, how do we find a matrix A such that $T(\vec{x}) = A\vec{x}$ for all $\vec{x} \in \mathbb{R}^n$?

1. Mathematical Foundations

The Identity Matrix (I_n)

The $n \times n$ identity matrix I_n has 1s on the main diagonal and 0s everywhere else:

$$I_n := \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ 0 & 0 & \ddots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

Property: for any vector $\vec{x} \in \mathbb{R}^n$, $I_n\vec{x} = \vec{x}$.

Example (3×3 case):

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Standard Basis Vectors

The columns of I_n form the standard basis vectors $\vec{e}_1, \vec{e}_2, \ldots, \vec{e}_n \in \mathbb{R}^n$:

$$\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \qquad \vec{e}_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \qquad \ldots, \qquad \vec{e}_n = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

Any vector

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n$$

can be expressed as a unique linear combination:

$$\vec{x} = x_1\vec{e}_1 + x_2\vec{e}_2 + \cdots + x_n\vec{e}_n$$

2. Construction of the Standard Matrix

Applying T to $\vec{x}$ and exploiting linearity ($T(c\vec{u} + d\vec{v}) = c\,T(\vec{u}) + d\,T(\vec{v})$):

$$\begin{aligned} T(\vec{x}) &= T(x_1\vec{e}_1 + x_2\vec{e}_2 + \cdots + x_n\vec{e}_n) \\ &= x_1\,T(\vec{e}_1) + x_2\,T(\vec{e}_2) + \cdots + x_n\,T(\vec{e}_n) \\ &= \left[\begin{array}{cccc} T(\vec{e}_1) & T(\vec{e}_2) & \cdots & T(\vec{e}_n) \end{array}\right] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \end{aligned}$$

Definition 7.3 (Standard matrix)**:** Given a linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$, the **standard matrix** associated with T is:

$$A = \left[\begin{array}{cccc} T(\vec{e}_1) & T(\vec{e}_2) & \cdots & T(\vec{e}_n) \end{array}\right]_{m\times n}$$

- **Key Fact:** $T(\vec{x}) = A\vec{x}$ for all $\vec{x} \in \mathbb{R}^n$.
- **Conceptual Observation:** a linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ is completely determined by its action on the standard basis vectors $T(\vec{e}_1), T(\vec{e}_2), \ldots, T(\vec{e}_n)$.
- **Equivalence:**

$$\{\text{Linear Transformations } T : \mathbb{R}^n \to \mathbb{R}^m\} \iff \{m \times n \text{ Matrices}\}$$

7.8. Examples and Geometric Diagrams

Example 1: Projection onto the xy-plane ($T : \mathbb{R}^3 \to \mathbb{R}^3$)

Find the standard matrix for T that projects 3D points perpendicularly onto the xy-plane.

Standard Basis Mapping

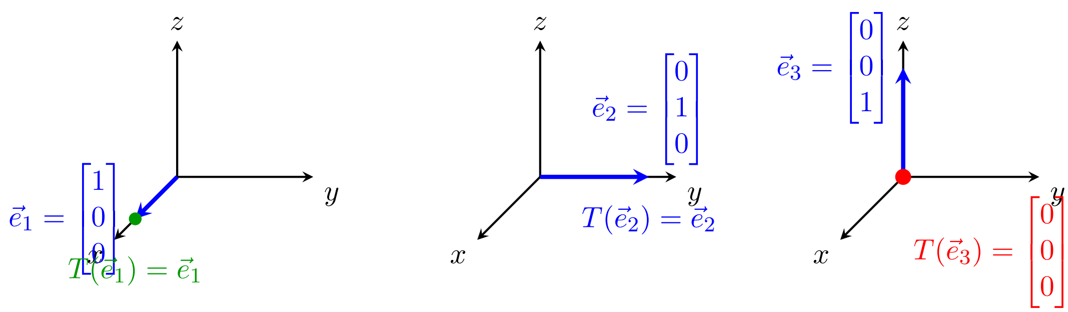
1. $\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \implies T(\vec{e}_1) = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \vec{e}_1$ (already on the xy-plane)
2. $\vec{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \implies T(\vec{e}_2) = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \vec{e}_2$ (already on the xy-plane)
3. $\vec{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \implies T(\vec{e}_3) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ (collapses to the origin)

Resulting Standard Matrix:

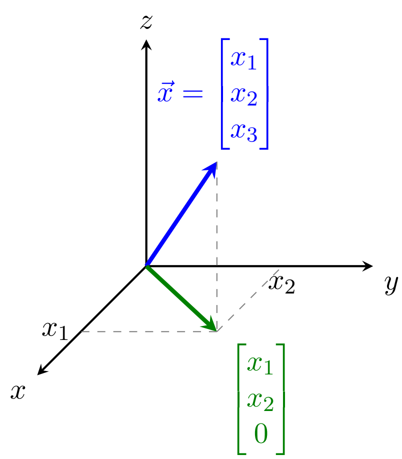
$$A = \left[\ T(\vec{e}_1) \quad T(\vec{e}_2) \quad T(\vec{e}_3)\ \right] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Reality Check:

$$A\vec{x} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ 0 \end{bmatrix}$$



Geometric Projection Reality Check



Example 2: 2D Counterclockwise Rotation ($T : \mathbb{R}^2 \to \mathbb{R}^2$)

Find the standard matrix for T rotating vectors counterclockwise about the origin by angle θ.

Standard Basis Mapping (Unit Circle Trigonometry)

1. **Action on** $\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$:

   Rotating by θ yields a unit vector at angle θ:

$$T(\vec{e}_1) = \begin{bmatrix} \cos(\theta) \\ \sin(\theta) \end{bmatrix}$$

2. **Action on** $\vec{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$:

   Rotating by θ shifts the position into the second quadrant (at angle $\theta + 90°$):

$$T(\vec{e}_2) = \begin{bmatrix} -\sin(\theta) \\ \cos(\theta) \end{bmatrix}$$

Resulting Standard Matrix:

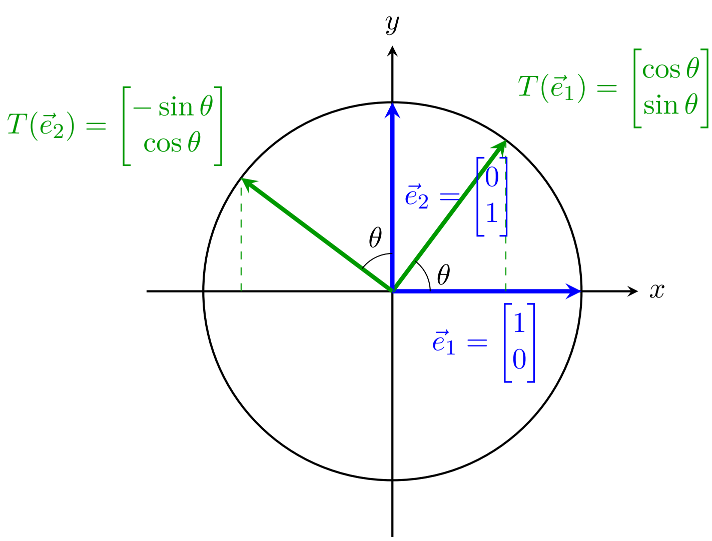
$$A = \begin{bmatrix} T(\vec{e}_1) & T(\vec{e}_2) \end{bmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

Trigonometric Triangle Reference

For a right triangle with hypotenuse 1 and angle θ:

$$\cos(\theta) = x/1 \implies x = \cos(\theta)$$

$$\sin(\theta) = y/1 \implies y = \sin(\theta)$$



7.9. Onto and One-to-One Linear Transformations

Linear transformations can be classified based on their mapping properties: whether they map onto the entire target space (*onto*) and whether distinct inputs map to distinct outputs (*one-to-one*).

1. Matrix Transformations: Domain, Codomain, and Range

Given an $m \times n$ matrix A, it defines a linear transformation:

$$T_A : \mathbb{R}^n \to \mathbb{R}^m, \qquad \vec{x} \mapsto A\vec{x}$$

- **Domain:** $\mathbb{R}^n$ (input space)
- **Codomain:** $\mathbb{R}^m$ (target space of all potential outputs)

- **Range of A (or T_A):** the set of all achieved outputs under T_A.

$$\text{Range of } A = \text{Image of } \mathbb{R}^n \text{ under } T_A = \text{span}\{\vec{a}_1, \dots, \vec{a}_n\}$$

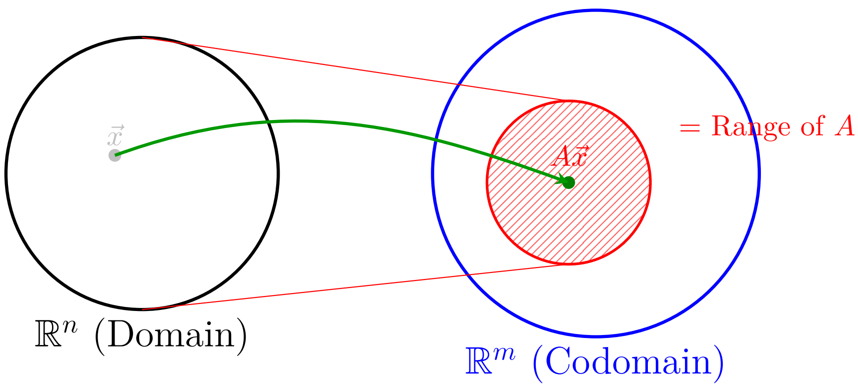
(where $\vec{a}_1, \dots, \vec{a}_n$ are the columns of matrix A, representing all linear combinations of $\vec{a}_1, \dots, \vec{a}_n$.)

Key Distinction: Codomain vs. Range

Codomain = all potential outputs
Range = all achieved outputs

(Note: the codomain and range are not always equal.)

Geometric Representation of Domain, Codomain, and Range



Onto Linear Transformations

Definition 7.4 (Onto)**:** A linear transformation $T_A : \mathbb{R}^n \to \mathbb{R}^m$ is **onto** if its range equals its codomain:

$$T_A \text{ is onto} \iff \text{Range of } T_A = \mathbb{R}^m \quad \text{(i.e., Range = Codomain)}$$

Equivalent Conditions for T_A Being Onto

$$T_A \text{ is onto} \iff \text{span}\{\vec{a}_1, \dots, \vec{a}_n\} = \mathbb{R}^m \iff [\,A \mid \vec{b}\,] \text{ is consistent for all } \vec{b} \in \mathbb{R}^m$$

$$\Updownarrow$$

A has **pivots in each row**

One-to-One Linear Transformations

Definition 7.5 (One-to-one)**:** A linear transformation $T_A : \mathbb{R}^n \to \mathbb{R}^m$ is **one-to-one (injective)** if distinct inputs map to distinct outputs:

$$\textbf{Def:} \quad T_A \text{ is one-to-one} \iff \text{if } T(\vec{u}) = T(\vec{v}),\ \text{then } \vec{u} = \vec{v}$$

Contrapositive Statement. For a conditional statement "If p, then q":

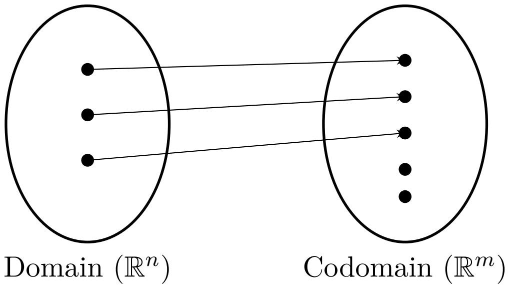
$$\text{Contrapositive: if not } q,\ \text{then not } p$$

Applying this to one-to-one transformations:

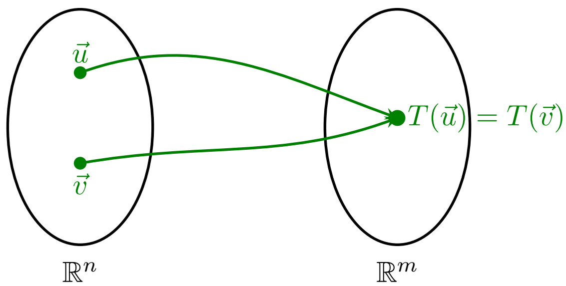
$$\text{Contrapositive: if } \vec{u} \neq \vec{v},\ \text{then } T(\vec{u}) \neq T(\vec{v})$$

Geometric Concepts: 1-1 vs. Not 1-1

One-to-One (1-1) Mapping. All distinct inputs produce distinct outputs:



Not One-to-One Mapping. Two or more distinct inputs yield the same output (e.g., $y = \sin\theta$):



Equivalent Conditions for T_A Being One-to-One

$$T_A \text{ is one-to-one} \iff \text{when } [\, A \mid \vec{b}\,] \text{ is consistent, there is a unique solution}$$

$$\Updownarrow$$

$$\text{Reduced } A \text{ has a pivot in each column}$$

$$\Updownarrow$$

$\{\vec{a}_1, \dots, \vec{a}_n\}$ are **Linearly Independent (L.I.)**

7.10. Examples: Testing Onto and One-to-One

Example 1: Testing Onto and One-to-One via Pivots

Let

$$A = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix}$$

1. Is T_A onto?
2. Is T_A one-to-one?

Solution: Perform Gaussian elimination to find pivot positions:

$$\begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix} \xrightarrow{-2R_1+R_2 \to R_2} \begin{bmatrix} 1 & 3 & 5 \\ 0 & -2 & -4 \end{bmatrix}$$

Pivot Positions: Row 1, Column 1 (1) and Row 2, Column 2 (-2).

Analysis:

1. **Onto Check:** there are 2 rows and every row contains a pivot position.

$$\text{Pivots in each row} \implies T_A \text{ is } \textbf{onto}$$

2. **One-to-One Check:** there are 3 columns, but only 2 columns contain pivot positions (Column 3 is a free variable column).

$$\text{Not all columns have pivots} \implies T_A \text{ is } \textbf{NOT} \text{ one-to-one}$$

Example 2: Analyzing Transformation Inclusivity from Matrix Multiplications

Suppose

$$A\begin{bmatrix}1\\3\\6\end{bmatrix}=\begin{bmatrix}1\\1\\1\end{bmatrix} \qquad \text{and} \qquad A\begin{bmatrix}2\\4\\8\end{bmatrix}=\begin{bmatrix}2\\2\\2\end{bmatrix}$$

Is T_A one-to-one? Answer using "definitely," "sometimes," or "definitely not." Explain.

Solution: Observe the scaling property of matrix–vector multiplication:

$$A\left(2\begin{bmatrix}1\\3\\6\end{bmatrix}\right)=2\left(A\begin{bmatrix}1\\3\\6\end{bmatrix}\right)=2\begin{bmatrix}1\\1\\1\end{bmatrix}=\begin{bmatrix}2\\2\\2\end{bmatrix}$$

This means:

$$A\begin{bmatrix}2\\6\\12\end{bmatrix}=\begin{bmatrix}2\\2\\2\end{bmatrix}$$

We now have two distinct input vectors that produce the exact same output:

$$\vec{u}=\begin{bmatrix}2\\4\\8\end{bmatrix} \qquad \text{and} \qquad \vec{v}=\begin{bmatrix}2\\6\\12\end{bmatrix} \qquad \implies \qquad \vec{u}\neq\vec{v}$$

However, their outputs under T_A are equal:

$$T_A(\vec{u})=A\vec{u}=\begin{bmatrix}2\\2\\2\end{bmatrix}=A\vec{v}=T_A(\vec{v})$$

Conclusion: there are two different inputs with the same output, so T_A is definitely not 1-1.

7.11. Fill-in-the-Blanks Review

The review below is typeset from the author's original LaTeX source, preserved exactly as provided:
Def: a function $T : \mathbb{R}^n \to \mathbb{R}^m$ is a linear transformation if it satisfies

1. $T(\boldsymbol{u} + \boldsymbol{v}) = T(\boldsymbol{u}) + T(\boldsymbol{v})$ for all $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$
2. $T(c\boldsymbol{u}) = c\,T(\boldsymbol{u})$ for all $\boldsymbol{u} \in \mathbb{R}^n$ and $c \in \mathbb{R}$

Def: we say a function $T : \mathbb{R}^n \to \mathbb{R}^m$ is **one-to-one (injective)** if $T(u) = T(v)$ implies $u = v$ for all $u,\ v \in \mathbb{R}^n$.

This is equivalent to saying whenever $u \neq v$, then ________________.

A function T is **NOT** one-to-one if there are two inputs $u \neq v$ such that ________________.

Def: we say a function $T : \mathbb{R}^n \to \mathbb{R}^m$ is **onto (surjective)** if $T(x) = b$ has a solution $x \in \mathbb{R}^n$ for all $b \in \mathbb{R}^m$.

A function $T : \mathbb{R}^n \to \mathbb{R}^m$ is **NOT** onto if there is ____________ such that ___________________.

Completed Statements

- This is equivalent to saying whenever $u \neq v$, then $T(u) \neq T(v)$.
- A function T is **NOT** one-to-one if there are two inputs $u \neq v$ such that $T(u) = T(v)$.
- A function $T : \mathbb{R}^n \to \mathbb{R}^m$ is **NOT** onto if there is **some** $b \in \mathbb{R}^m$ such that $T(x) = b$ has **no solution** (or b is not in the range of T).

7.12. Example Question: Rank–Nullity Reasoning

Example 7.2: Suppose $T : \mathbb{R}^3 \to \mathbb{R}^2$ is linear. Which of the following statements would imply that T is onto? Briefly explain. There may be more than one right answer.

- **(a)** $T(\vec{x}) = \vec{b}$ has infinitely many solutions for every $\vec{b} \in \mathbb{R}^2$
- **(b)** $T(\vec{x}) = [\,4 \quad 2\,]$ is consistent
- **(c)** The solution set to $T(\vec{x}) = [\,0 \quad 0\,]$ is a line through the origin

Answer & Explanations:

- **(a) Implies T is onto.**

  Reason: the definition of an onto transformation requires that $T(\vec{x}) = \vec{b}$ has *at least one* solution for every $\vec{b} \in \mathbb{R}^2$. Having infinitely many solutions for every $\vec{b}$ guarantees that every target vector is mapped to, satisfying the definition completely.

- **(b) Does NOT imply T is onto.**

  Reason: being consistent for a single *specific* vector

$$\begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

does not guarantee that $T(\vec{x}) = \vec{b}$ has a solution for *all* vectors $\vec{b} \in \mathbb{R}^2$.

- **(c) Implies T is onto.**

  Reason: by the Rank–Nullity Theorem:

$$\dim(\text{Domain}) = \dim(\text{Kernel}) + \operatorname{rank}(T)$$

  Here, the domain dimension is 3. The solution set to $T(\vec{x}) = \vec{0}$ (the kernel / null space) is a 1-dimensional line ($\dim(\text{Kernel}) = 1$). Thus:

$$3 = 1 + \operatorname{rank}(T) \quad \Longrightarrow \quad \operatorname{rank}(T) = 2$$

  Since the dimension of the range is 2 (which equals the dimension of the codomain $\mathbb{R}^2$), the range of T spans all of $\mathbb{R}^2$, meaning T is onto.

Correct Options: (a) and (c)

END OF THE BOOK · The series continues with the next topics: Matrix operations, vector spaces, etc.